

Part A · Multiple Choice

2 points each

1. **a) Hydrolysis of feldspar into clay minerals**

p. 165 — hydrolysis is a chemical weathering process. The fourth mechanical agent is cracking by plant roots and exposure by burrowing animals.

2. **b) freezing and thawing alternate frequently**

p. 166 — its effectiveness depends on the frequency of freeze–thaw cycles. Talus slopes at the base of steep rocky slopes are the characteristic result.

3. **b) clay minerals such as kaolinite**

p. 170 — along with dissolved calcium and carbonate ions, which may later combine in the ocean to form calcite. Pyroxene similarly alters to chlorite or smectite, and olivine to serpentine.

4. **a) water combines with carbon dioxide**

p. 169 — $H_2O + CO_2 \rightarrow H_2CO_3$. Soil typically holds far more CO_2 than the air, so water percolating through soil becomes significantly more acidic.

5. **d) About 50%**

p. 174 — the rest is about 45% minerals and about 5% organic matter.

6. **c) E**

p. 177 — the horizons run O (organic matter), A (partially decayed organics mixed with mineral material), E (leached), B (accumulation of clay and iron from above), C (incomplete weathering), and R (parent material or bedrock).

Part B · True or False

1 point each

7. **TRUE**

p. 169 — which is exactly why quartz grains dominate sand and are a principal product of weathering and erosion. Feldspar, by contrast, is easily altered.

8. **FALSE**

Soil can only develop where surface material stays in place. On steep slopes the rate of erosion exceeds the rate of soil formation, so steep slopes tend to have little or no soil. (p. 176)

Part C · Short Answer

3 points each

9. **Mechanical weathering breaks rock into smaller fragments, exposing fresh surfaces for chemical attack. Chemical weathering alters minerals and weakens the rock, making it more susceptible to mechanical breakdown.**

Rubric: 1.5 pts each direction. **Mechanical helps chemical** — breaking rock into smaller pieces creates far more surface area, and every new surface is fresh ground for chemical processes to attack. **Chemical helps mechanical** — altering minerals to less stable forms weakens the rock so it fractures more readily. Credit the note that together they yield two critical products: sedimentary clasts and ions that become sedimentary rock, and the soil we depend on. (p. 165)

10. Climate, parent material, slope, and the amount of time available.

Rubric: 2 pts for naming all four (0.5 each). 1 pt for a correct explanation of any one. **Climate** — soil forms best under temperate to tropical conditions with moderate precipitation; too much water leaches nutrients and acidifies, too little leaves salts and carbonates and little organic matter. **Parent material** — quartz-rich rock like granite yields sandy soil, while basalt yields fertile soil rich in phosphorus, iron, magnesium, and calcium. **Slope** — steep slopes erode faster than soil can form. **Time** — even under ideal conditions soil takes thousands of years, so recently glaciated ground has young, poorly developed soil. (pp. 174–176)